



Exercise-1

Marked questions are recommended for Revision.

PART - I : SUBJECTIVE QUESTIONS

Section (A) : Integration using Standard Integral :

A-1. Integrate with respect to x :

- | | | |
|------------------------|---------------------|-------------------------|
| (i) $(2x + 3)^5$ | (ii) $\sin 2x$ | (iii) $\sec^2 (4x + 5)$ |
| (iv) $\sec (3x + 2)$ | (v) $\tan (2x + 1)$ | (vi) 2^{3x+4} |
| (vii) $\frac{1}{2x+1}$ | (viii) e^{4x+5} | |

A-2. Integrate with respect to x :

- | | | |
|--|---|-------------------------|
| (i) $\sin^2 x$ | (ii) $\cos^3 x$ | (iii) $\sin 2x \cos 3x$ |
| (iv) $4 \sin x \cos \frac{x}{2} \cos \frac{3x}{2}$ | (v) $\frac{1}{\sqrt{x+3} - \sqrt{x+2}}$ | |

Section (B) : Integration using Substitution :

B-1. Integrate with respect to x :

- | | | | |
|---|--|--|---|
| (i) $x \sin x^2$ | (ii) $\frac{x}{x^2+1}$ | (iii) $\sec^2 x \tan x$ | (iv) $\frac{e^x+1}{e^x+x}$ |
| (v) $\frac{1-\sin x}{x+\cos x}$ | (vi) $\frac{e^{2x}}{e^{2x}-2}$ | (vii) $\frac{\cos 2x + x + 1}{x^2 + \sin 2x + 2x}$ | (viii) $\frac{\sec x}{\ln (\sec x + \tan x)}$ |
| (ix) $\frac{x}{\sqrt{x+2}}$ | (x) $\left(e^x + \frac{1}{e^x}\right)^2$ | (xi) $(e^x + 1)^2 e^x$ | (xii) $\frac{1}{x(x^5+1)}$ |
| (xiii) $\frac{1}{x^5(1+x^5)^{\frac{1}{5}}}$ | (xiv) $\frac{\sqrt{x^2-8}}{x^4}$ | | |

B-2. Find the value of $\int \frac{d(x^2+1)}{\sqrt{(x^2+2)}}$.

B-3. Evaluate the following :

- | | |
|---|--|
| (i) $\int \left(\frac{x \cos x - \sin x}{x \sin x} \right) dx$ | (ii) $\int \left(\frac{\frac{x}{x+1} - \ln(x+1)}{x(\ln(x+1))} \right) dx$ |
|---|--|

Section (C) : Integration by parts :

C-1. Integrate with respect to x :

- | | | | |
|-------------------|-------------------------------|----------------------------|--|
| (i) $x \ln x$ | (ii) $x \sin^2 x$ | (iii) $x \tan^{-1} x$ | (iv) $\ln x$ |
| (v) $\sec^3 x$ | (vi) $2x^3 e^{x^2}$ | (vii) $\sin^{-1} \sqrt{x}$ | (viii) $\frac{x^2 \tan^{-1} x}{1+x^2}$ |
| (ix) $e^x \sin x$ | (x) $e^x (\sec^2 x + \tan x)$ | | |

C-2. Find the antiderivative of $f(x) = \ln (\ln x) + (\ln x)^{-2}$ whose graph passes through (e, e) .





Section (D) : Algebraic integral :

D-1. Integrate with respect to x :

(i) $\frac{1}{x^2 + 4}$

(ii) $\frac{1}{x^2 + 5}$

(iii) $\frac{1}{x^2 + 2x + 5}$

(iv) $\frac{2x + 1}{x^2 + 3x + 4}$

(v) $\frac{x^3 - 1}{x^3 + x}$

(vi) $\frac{1}{\sqrt{x^2 - 4}}$

(vii) $\sqrt{x^2 + 4}$

(viii) $\sqrt{x^2 + 2x + 5}$

(ix) $(x - 1) \sqrt{1 - x - x^2}$

(x) $x^5 \sqrt{a^3 + x^3}$

D-2. Integrate with respect to x :

(i) $\frac{1}{(x + 1)(x + 2)}$

(ii) $\frac{1}{(x^2 + 1)(x + 3)}$

(iii) $\frac{3x + 2}{(x + 1)^2(x + 2)}$

(iv) $\frac{1}{(x + 1)(x + 2)(x + 3)}$

D-3. Integrate with respect to x :

(i) $\frac{1}{x^4 + x^2 + 1}$

(ii) $\frac{1 + x^2}{1 + x^4}$

(iii) $\frac{1 - x^2}{1 - x^2 + x^4}$

D-4. Integrate with respect to x :

(i) $\frac{1}{(x + 1)\sqrt{x + 2}}$

(ii) $\frac{1}{(x^2 - 4)\sqrt{x + 1}}$

(iii) $\frac{1}{(x + 1)\sqrt{x^2 + 2}}$

(iv) $\frac{1}{(x^2 + 1)\sqrt{x^2 + 2}}$

D-5. Evaluate the following :

(i) $\int \sqrt{\frac{1+x}{x}} dx$

(ii) $\int \sqrt{\frac{x-1}{x+1}} dx$

(iii) $\int \left(\frac{x\sqrt{1+x}}{\sqrt{1-x}} \right) dx$

Section (E) : Integration of trigonometric functions :

E-1. Integrate with respect to x :

(i) $\frac{1}{2 + \cos x}$

(ii) $\frac{1}{2 - \cos x}$

(iii) $\frac{2\sin x + 2\cos x}{3\cos x + 2\sin x}$

(iv) $\frac{1}{1 + \sin x + \cos x}$

(v) $\frac{1}{2 + \sin^2 x}$

(vi) $\frac{\operatorname{cosec}^2 x \cdot \sin x}{(\sin x - \cos x)}$

(vii) $\frac{\sin^4 x}{\cos^2 x}$

E-2. Evaluate the following

(i) $\int \left(\frac{\sin x + \cos x}{9 + 16 \sin 2x} \right) dx$

(ii) $\int \left(\frac{\cos x - \sin x}{\sqrt{8 - \sin 2x}} \right) dx$

E-3. If $\int \sqrt{\frac{\cos^3 x}{\sin^{11} x}} dx = -2 \left(A \tan^{\frac{-9}{2}} x + B \tan^{\frac{-5}{2}} x \right) + C$, then find A and B .



Section (F) : Reduction formulae

F-1. If $I_n = \int \frac{1}{(x^2 + a^2)^n} dx$ then prove that $I_n = \frac{x}{2a^2(n-1)(x^2 + a^2)^{n-1}} + \frac{2n-3}{2(n-1)a^2} I_{n-1}$

F-2. If $I_n = \int x^n (a-x)^{1/2} dx$ then prove that $I_n = \frac{2an}{2n+3} I_{n-1} - \frac{2x^n(a-x)^{3/2}}{2n+3}$

PART - II : ONLY ONE OPTION CORRECT TYPE

Section (A) : Integration using Standard Integral :

A-1. Integrate with respect to $x : \sqrt{x+1}$
 (A) $\frac{(x+1)^{3/2}}{2} + C$ (B) $\frac{3(x+1)^{3/2}}{2} + C$ (C) $\frac{(x+1)^{3/2}}{3} + C$ (D) $\frac{2(x+1)^{3/2}}{3} + C$

A-2. Integrate with respect to $x : \frac{1}{\sqrt{2x+1}}$
 (A) $\sqrt{2x+1} + C$ (B) $(2x+1)^{3/2} + C$ (C) $-\sqrt{2x+1} + C$ (D) $\frac{1}{(2x+1)^{3/2}} + C$

A-3. If $\int \frac{1}{1+\sin x} dx = \tan\left(\frac{x}{2} + a\right) + C$, then
 (A) $a = -\frac{\pi}{4}, C \in \mathbb{R}$ (B) $a = \frac{\pi}{4}, C \in \mathbb{R}$ (C) $a = \frac{5\pi}{4}, C \in \mathbb{R}$ (D) $a = \frac{\pi}{3}, C \in \mathbb{R}$

A-4. If $\int (\sin 2x - \cos 2x) dx = \frac{1}{\sqrt{2}} \sin(2x - a) + C$, then
 (A) $a = \frac{5\pi}{4}, C \in \mathbb{R}$ (B) $a = -\frac{5\pi}{4}, C \in \mathbb{R}$ (C) $a = \frac{\pi}{4}, C \in \mathbb{R}$ (D) $a = \frac{\pi}{2}, C \in \mathbb{R}$

A-5. The value of $\int \frac{\cos 2x}{\cos x} dx$ is equal to
 (A) $2 \sin x - \ln |\sec x + \tan x| + C$ (B) $2 \sin x - \ln |\sec x - \tan x| + C$
 (C) $2 \sin x + \ln |\sec x + \tan x| + C$ (D) $\sin x - \ln |\sec x - \tan x| + C$

A-6. If $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx = A \cos 4x + B$; where A & B are constants, then
 (A) $A = -1/4$ & B may have any value (B) $A = -1/8$ & B may have any value
 (C) $A = -1/2$ & $B = -1/4$ (D) $A = B = 1/2$

Section (B) : Integration using Substitution :

B-1. The value of $\int \frac{a^{\sqrt{x}}}{\sqrt{x}} dx$ is equal to
 (A) $\frac{a^{\sqrt{x}}}{\sqrt{x}} + C$ (B) $\frac{2a^{\sqrt{x}}}{\ln a} + C$ (C) $2a^{\sqrt{x}} \cdot \ln a + C$ (D) $2a^{\sqrt{x}} + C$

B-2. The value of $\int 5^{5^x} \cdot 5^x \cdot 5^x dx$ is equal to
 (A) $\frac{5^{5^x}}{(\ln 5)^3} + C$ (B) $5^{5^x} (\ln 5)^3 + C$ (C) $\frac{5^{5^x}}{(\ln 5)^3} + C$ (D) $\frac{5^{5^x}}{(\ln 5)^2} + C$



B-3. The value of $\int \frac{\sqrt{\tan x}}{\sin x \cos x} dx$ is equal to

- (A) $2\sqrt{\tan x} + C$ (B) $2\sqrt{\cot x} + C$ (C) $\frac{\sqrt{\tan x}}{2} + C$ (D) $\sqrt{\tan x} + C$

B-4. If $\int \frac{2^x}{\sqrt{1-4^x}} dx = K \sin^{-1}(2^x) + C$, then the value of K is equal to

- (A) $\ln 2$ (B) $\frac{1}{2} \ln 2$ (C) $\frac{1}{2}$ (D) $\frac{1}{\ln 2}$

B-5. If $y = \int \frac{dx}{(1+x^2)^{3/2}}$ and $y = 0$ when $x = 0$, then value of y when $x = 1$, is:

- (A) $\sqrt{\frac{2}{3}}$ (B) $\sqrt{2}$ (C) $3\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

B-6. The value of $\int \tan^3 2x \sec 2x dx$ is equal to :

- (A) $\frac{1}{3} \sec^3 2x - \frac{1}{2} \sec 2x + C$ (B) $-\frac{1}{6} \sec^3 2x - \frac{1}{2} \sec 2x + C$
 (C) $\frac{1}{6} \sec^3 2x - \frac{1}{2} \sec 2x + C$ (D) $\frac{1}{3} \sec^3 2x + \frac{1}{2} \sec 2x + C$

B-7. If $\int x^{13/2} \cdot (1+x^{5/2})^{1/2} dx = P(1+x^{5/2})^{7/2} + Q(1+x^{5/2})^{5/2} + R(1+x^{5/2})^{3/2} + C$, then P, Q and R are

- (A) $P = \frac{4}{35}$, $Q = -\frac{8}{25}$, $R = \frac{4}{15}$ (B) $P = \frac{4}{35}$, $Q = \frac{8}{25}$, $R = \frac{4}{15}$
 (C) $P = -\frac{4}{35}$, $Q = -\frac{8}{25}$, $R = \frac{4}{15}$ (D) $P = \frac{4}{35}$, $Q = -\frac{8}{25}$, $R = -\frac{4}{15}$

B-8. The value of $\int \frac{1-x^7}{x(1+x^7)} dx$ is equal to

- (A) $\ln|x| + \frac{2}{7} \ln|1+x^7| + C$ (B) $\ln|x| - \frac{2}{7} \ln|1-x^7| + C$
 (C) $\ln|x| - \frac{2}{7} \ln|1+x^7| + C$ (D) $\ln|x| + \frac{2}{7} \ln|1-x^7| + C$

Section (C) : Integration by parts :

C-1. The value of $\int (x-1) e^{-x} dx$ is equal to

- (A) $-xe^x + C$ (B) $xe^x + C$ (C) $-xe^{-x} + C$ (D) $xe^{-x} + C$

C-2. The value of $\int e^{\tan^{-1} x} \left(\frac{1+x+x^2}{1+x^2} \right) dx$ is equal to

- (A) $xe^{\tan^{-1} x} + C$ (B) $x^2 e^{\tan^{-1} x} + C$ (C) $\frac{1}{x} e^{\tan^{-1} x} + C$ (D) $xe^{\cot^{-1} x} + C$

C-3. The value of $\int [f(x)g''(x) - f''(x)g(x)] dx$ is equal to

- (A) $\frac{f(x)}{g'(x)} + C$ (B) $f'(x)g(x) - f(x)g'(x) + C$
 (C) $f(x)g'(x) - f'(x)g(x) + C$ (D) $f(x)g'(x) + f'(x)g(x) + C$



C-4. $\int \frac{x \ln x}{(x^2 - 1)^{3/2}} dx$ equals

(A) $\arcsin x - \frac{\ln x}{\sqrt{x^2 - 1}} + C$

(B) $\sec^{-1} x + \frac{\ln x}{\sqrt{x^2 - 1}} + C$

(C) $\cos^{-1} x - \frac{\ln x}{\sqrt{x^2 - 1}} + C$

(D) $\sec x - \frac{\ln x}{\sqrt{x^2 - 1}} + C$

C-5. The value of $\int (x e^{\ln \sin x} - \cos x) dx$ is equal to:

(A) $x \cos x + C$

(B) $\sin x - x \cos x + C$

(C) $-e^{\ln x} \cos x + C$

(D) $\sin x + x \cos x + C$

Section (D) : Algebraic integral :

D-1. The value of $\int \frac{dx}{x^2 + x + 1}$ is equal to

(A) $\frac{\sqrt{3}}{2} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$

(B) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$

(C) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + C$

(D) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) + C$

D-2. The value of $\int \frac{1}{x^2(x^4 + 1)^{3/4}} dx$ is equal to

(A) $\left(1 + \frac{1}{x^4}\right)^{1/4} + C$

(B) $(x^4 + 1)^{1/4} + C$

(C) $\left(1 - \frac{1}{x^4}\right)^{1/4} + C$

(D) $-\left(1 + \frac{1}{x^4}\right)^{1/4} + C$

D-3. The value of $\int \frac{dx}{x\sqrt{1-x^3}}$ is equal to

(A) $\frac{1}{3} \ln \left| \frac{\sqrt{1-x^3}-1}{\sqrt{1-x^3}+1} \right| + C$

(B) $\frac{1}{3} \ln \left| \frac{\sqrt{1-x^2}+1}{\sqrt{1-x^2}-1} \right| + C$

(C) $\frac{1}{3} \ln \left| \frac{1}{\sqrt{1-x^3}} \right| + C$

(D) $\frac{1}{3} \ln |1-x^3| + C$

D-4. The value of $\int \sqrt{\frac{e^x-1}{e^x+1}} dx$ is equal to

(A) $\ln(e^x + \sqrt{e^{2x}-1}) - \sec^{-1}(e^x) + C$

(B) $\ln(e^x + \sqrt{e^{2x}-1}) + \sec^{-1}(e^x) + C$

(C) $\ln(e^x - \sqrt{e^{2x}-1}) - \sec^{-1}(e^x) + C$

(D) $\ln(e^x + \sqrt{e^{2x}-1}) - \sin^{-1}(e^x) + C$

D-5. If $\int \frac{dx}{x^4 + x^3} = \frac{A}{x^2} + \frac{B}{x} + \ln \left| \frac{x}{x+1} \right| + C$, then

(A) $A = \frac{1}{2}, B = 1$

(B) $A = 1, B = -\frac{1}{2}$

(C) $A = -\frac{1}{2}, B = 1$

(D) $A = -\frac{1}{2}, B = \frac{1}{2}$

Section (E) : Integration of trigonometric functions :

E-1. The value of $\int \frac{\cos 2x}{(\sin x + \cos x)^2} dx$ is equal to

(A) $\frac{-1}{\sin x + \cos x} + C$

(B) $\ln(\sin x + \cos x) + C$

(C) $\ln(\sin x - \cos x) + C$

(D) $\ln(\sin x + \cos x)^2 + C$



E-2 The value of $\int [1 + \tan x \cdot \tan(x + \alpha)] dx$ is equal to

- (A) $\cos \alpha \cdot \ln \left| \frac{\sin x}{\sin(x + \alpha)} \right| + C$ (B) $\tan \alpha \cdot \ln \left| \frac{\sin x}{\sin(x + \alpha)} \right| + C$
 (C) $\cot \alpha \cdot \ln \left| \frac{\sec(x + \alpha)}{\sec x} \right| + C$ (D) $\cot \alpha \cdot \ln \left| \frac{\cos(x + \alpha)}{\cos x} \right| + C$

E-3. The value of $\int \sqrt{\sec x - 1} dx$ is equal to

- (A) $2 \ln \left(\cos \frac{x}{2} + \sqrt{\cos^2 \frac{x}{2} - \frac{1}{2}} \right) + C$ (B) $\ln \left(\cos \frac{x}{2} + \sqrt{\cos^2 \frac{x}{2} - \frac{1}{2}} \right) + C$
 (C) $-2 \ln \left(\cos \frac{x}{2} + \sqrt{\cos^2 \frac{x}{2} - \frac{1}{2}} \right) + C$ (D) $-2 \ln \left(\sin \frac{x}{2} + \sqrt{\cos^2 \frac{x}{2} - \frac{1}{2}} \right) + C$

E-4. The value of $\int \frac{dx}{\cos^3 x \sqrt{\sin 2x}}$ is equal to

- (A) $\sqrt{2} \left(\sqrt{\cos x} + \frac{1}{5} \tan^{5/2} x \right) + C$ (B) $\sqrt{2} \left(\sqrt{\tan x} + \frac{1}{5} \tan^{5/2} x \right) + C$
 (C) $\sqrt{2} \left(\sqrt{\tan x} - \frac{1}{5} \tan^{5/2} x \right) + C$ (D) $\sqrt{2} \left(\sqrt{\cos x} - \frac{1}{5} \tan^{5/2} x \right) + C$

E-5. Antiderivative of $\frac{\sin^2 x}{1 + \sin^2 x}$ w.r.t. x is :

- (A) $x - \frac{\sqrt{2}}{2} \arctan(\sqrt{2} \tan x) + C$ (B) $x - \frac{1}{\sqrt{2}} \arctan\left(\frac{\tan x}{\sqrt{2}}\right) + C$
 (C) $x - \sqrt{2} \arctan(\sqrt{2} \tan x) + C$ (D) $x - \sqrt{2} \arctan\left(\frac{\tan x}{\sqrt{2}}\right) + C$

E-6. Integrate $\frac{1}{1 - \cot x}$

- (A) $\frac{1}{2} \log |\sin x - \cos x| + \frac{1}{2} x + C$ (B) $\frac{1}{2} \log |\sin x + \cos x| + \frac{1}{2} x + C$
 (C) $\frac{1}{2} \log |\sin x + \cos x| - \frac{1}{2} x + C$ (D) $\frac{1}{2} \log |\sin x - \cos x| - \frac{1}{2} x + C$

E-7. $I = \int \frac{dx}{\sin x + \sec x}$ is equal to

- (A) $\frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - (\sin x - \cos x)} \right| + \tan^{-1}(\sin x + \cos x) + C$
 (B) $\frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - (\sin x - \cos x)} \right| + \tan^{-1}(\sin x - \cos x) + C$
 (C) $\frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + \sin x + \cos x}{\sqrt{3} - (\sin x - \cos x)} \right| + \tan^{-1}(\sin x + \cos x) + C$
 (D) $\frac{1}{2\sqrt{3}} \log \left| \frac{\sqrt{3} + \sin x - \cos x}{\sqrt{3} - (\sin x + \cos x)} \right| + \tan^{-1}(\sin x + \cos x) + C$



Section (F) : Reduction formulae

F-1. If $I_n = \int \frac{e^x}{x^n} dx$ and $I_n = \frac{-e^x}{k_1 x^{n-1}} + \frac{1}{k_2 - 1} I_{n-1}$, then $(k_2 - k_1)$ is equal to :

- (A) 0 (B) 1 (C) 2 (D) 3

F-2. If $I_n = \int \cot^n x \, dx$ and $I_0 + I_1 + 2(I_2 + \dots + I_8) + I_9 + I_{10} = A \left(u + \frac{u^2}{2} + \dots + \frac{u^9}{9} \right) + C$, where $u = \cot x$ and C is an arbitrary constant, then

(A) $A = 2$ (B) $A = -1$ (C) $A = 1$ (D) A is dependent on x

PART - III : MATCH THE COLUMN

1. Column – I

Column – II

(A) If $F(x) = \int \frac{x + \sin x}{1 + \cos x} dx$ and $F(0) = 0$, then the value of $F(\pi/2)$ is (p) $\frac{\pi}{2}$

(B) Let $F(x) = \int e^{\sin^{-1} x} \left(1 - \frac{x}{\sqrt{1-x^2}} \right) dx$ and $F(0) = 1$, (q) $\frac{\pi}{3}$

If $F(1/2) = \frac{k\sqrt{3}}{\pi} e^{\pi/6}$, then the value of k is

(C) Let $F(x) = \int \frac{dx}{(x^2 + 1)(x^2 + 9)}$ and $F(0) = 0$, (r) $\frac{\pi}{4}$

if $F(\sqrt{3}) = \frac{5}{36} k$, then the value of k is

(D) Let $F(x) = \int \frac{\sqrt{\tan x}}{\sin x \cos x} dx$ and $F(0) = 0$ (s) π

if $F(\pi/4) = \frac{2k}{\pi}$, then the value of k is

2. If $I = \int \frac{dx}{a+b \cos x}$, where $a, b > 0$ and $a + b = u$, $a - b = v$, then match the following column

Column – I

Column – II

(A) $v = 0$

(p) $I = \frac{1}{\sqrt{uv}} \ln \left| \frac{\sqrt{u} + \sqrt{v} \tan \frac{x}{2}}{\sqrt{u} - \sqrt{v} \tan \frac{x}{2}} \right| + C$

(B) $v > 0$

(q) $I = \frac{2}{\sqrt{uv}} \tan^{-1} \left(\sqrt{\frac{v}{u}} \tan \frac{x}{2} \right) + C$

(C) $v < 0$

(r) $I = \frac{1}{\sqrt{-u} \, v} \ln \left| \frac{\sqrt{u} + \sqrt{-v} \tan \frac{x}{2}}{\sqrt{u} - \sqrt{-v} \tan \frac{x}{2}} \right| + C$

(s) $\frac{2}{u} \tan \frac{x}{2} + C$



Exercise-2

Marked questions are recommended for Revision.

PART - I : ONLY ONE OPTION CORRECT TYPE

- Value of $\int \frac{1}{\sin(x-a) \cos(x-b)} dx$ is equal to

(A) $\frac{1}{\cos(a-b)} \ln \left| \frac{\sin(x-a)}{\cos(x-b)} \right| + C$ (B) $\frac{1}{\cos(a-b)} \ln \left| \frac{\cos(x-b)}{\sin(x-a)} \right| + C$

(C) $\frac{1}{\sin(a-b)} \ln \left| \frac{\sin(x-a)}{\cos(x-b)} \right| + C$ (D) $\frac{1}{\sin(a+b)} \ln \left| \frac{\cos(x-a)}{\sin(x-b)} \right| + C$
- $\int \tan x \cdot \tan 2x \cdot \tan 3x dx =$

(A) $-\ln|\cos x| - \frac{1}{2} \ln|\sec 2x| + \frac{1}{3} \ln|\sec 3x| + C$

(B) $-\ln|\sec x| - \frac{1}{2} \ln|\sec 2x| + \frac{1}{3} \ln|\sec 3x| + C$

(C) $\ln|\cos x| + \ln|\cos 2x| + \ln|\cos 3x| + C$

(D) $\ln|\sec x| + \frac{1}{2} \ln|\sec 2x| + \frac{1}{3} \ln|\sec 3x| + C$
- The value of $\int (\sin x \cdot \cos x \cdot \cos 2x \cdot \cos 4x \cdot \cos 8x \cdot \cos 16x) dx$ is equal to

(A) $\frac{\sin 16x}{1024} + C$ (B) $-\frac{\cos 32x}{1024} + C$ (C) $\frac{\cos 32x}{1096} + C$ (D) $-\frac{\cos 32x}{1096} + C$
- $\int x \sqrt{\frac{a^2 - x^2}{a^2 + x^2}} dx =$

(A) $\frac{1}{2} a^2 \cos^{-1} \left(\frac{x^2}{a^2} \right) + \frac{1}{2} \sqrt{a^4 + x^4} + C$ (B) $\frac{1}{2} \sin^{-1} \left(\frac{x^2}{a^2} \right) + \sqrt{a^4 + x^4} + C$

(C) $\frac{1}{2} a^2 \sin^{-1} \left(\frac{x^2}{a^2} \right) + \frac{1}{2} \sqrt{a^4 - x^4} + C$ (D) $\frac{1}{2} \cos^{-1} \left(\frac{x^2}{a^2} \right) + \frac{1}{2} \sqrt{a^4 - x^4} + C$
- The value of $\int \sqrt{\frac{x-1}{x+1}} \cdot \frac{1}{x^2} dx$ is equal to

(A) $\sin^{-1} \frac{1}{x} + \frac{\sqrt{x^2-1}}{x} + C$ (B) $\frac{\sqrt{x^2-1}}{x} + \cos^{-1} \frac{1}{x} + C$

(C) $\sec^{-1} x - \frac{\sqrt{x^2-1}}{x} + C$ (D) $\tan^{-1} \sqrt{x^2+1} - \frac{\sqrt{x^2-1}}{x} + C$





6. The value of $\int \frac{\ell n |x|}{x \sqrt{1 + \ell n |x|}} dx$ equals :
- (A) $\frac{2}{3} \sqrt{1 + \ell n |x|} (\ell n |x| - 2) + C$ (B) $\frac{2}{3} \sqrt{1 + \ell n |x|} (\ell n |x| + 2) + C$
 (C) $\frac{1}{3} \sqrt{1 + \ell n |x|} (\ell n |x| - 2) + C$ (D) $2 \sqrt{1 + \ell n |x|} (3 \ell n |x| - 2) + C$
7. The value of $\int \frac{1}{[(x-1)^3(x+2)^5]^{1/4}} dx$ is equal to
- (A) $\frac{4}{3} \left(\frac{x-1}{x+2}\right)^{1/4} + C$ (B) $\frac{4}{3} \left(\frac{x+2}{x-1}\right)^{1/4} + C$ (C) $\frac{1}{3} \left(\frac{x-1}{x+2}\right)^{1/4} + C$ (D) $\frac{1}{3} \left(\frac{x+1}{x-1}\right)^{1/4} + C$
8. The value of $\int \frac{\sqrt{1-\sqrt{x}}}{1+\sqrt{x}} dx$ is equal to
- (A) $\sqrt{x} \sqrt{1-x} - 2 \sqrt{1-x} + \cos^{-1}(\sqrt{x}) + C$ (B) $\sqrt{x} \sqrt{1-x} + 2 \sqrt{1-x} + \cos^{-1}(\sqrt{x}) + C$
 (C) $\sqrt{x} \sqrt{1-x} - 2 \sqrt{1-x} - \cos^{-1}(\sqrt{x}) + C$ (D) $\sqrt{x} \sqrt{1-x} + 2 \sqrt{1-x} - \cos^{-1}(\sqrt{x}) + C$
9. $\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx$ is equal to
- (A) $(a+x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$ (B) $(a+x) \arctan \sqrt{\frac{x}{a}} + \sqrt{ax} + C$
 (C) $(a-x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$ (D) $(a+x) \arctan \sqrt{\frac{x}{a}} - \sqrt{ax} + C$
10. The value of $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} (x + \sqrt{x}) dx$ is equal to :
- (A) $2e^{\sqrt{x}} [\sqrt{x} - x + 1] + C$ (B) $2e^{\sqrt{x}} [x - 2\sqrt{x} + 1] + C$
 (C) $2e^{\sqrt{x}} [x - \sqrt{x} + 1] + C$ (D) $2e^{\sqrt{x}} (x + \sqrt{x} + 1) + C$
11. If $I = \int \frac{2}{x} (x^{\ell n x}) (\ell n x)^3 dx = Ax^{\ell n x} (\ell n x)^2 - Bx^{\ell n x} + C$, then $\frac{A}{B}$ is equal to :
- (A) 1 (B) -1 (C) 2 (D) -2
12. The value of $\int e^{\tan \theta} (\sec \theta - \sin \theta) d\theta$ is equal to
- (A) $-e^{\tan \theta} \sin \theta + C$ (B) $e^{\tan \theta} \sin \theta + C$ (C) $e^{\tan \theta} \sec \theta + C$ (D) $e^{\tan \theta} \cos \theta + C$
13. The value of $\int \left\{ \ln(1 + \sin x) + x \tan\left(\frac{\pi}{4} - \frac{x}{2}\right) \right\} dx$ is equal to:
- (A) $x \ell n(1 + \sin x) + C$ (B) $\ell n(1 + \sin x) + C$ (C) $-x \ell n(1 + \sin x) + C$ (D) $\ell n(1 - \sin x) + C$
14. The value of $\int x \cdot \frac{\ell n(x + \sqrt{1+x^2})}{\sqrt{1+x^2}} dx$ equals:
- (A) $\sqrt{1+x^2} \ell n(x + \sqrt{1+x^2}) - x + C$ (B) $\frac{x}{2} \cdot \ell n^2(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$
 (C) $\frac{x}{2} \cdot \ell n^2(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + C$ (D) $\sqrt{1+x^2} \ell n(x + \sqrt{1+x^2}) + x + C$



15. If $\int \frac{x \tan^{-1} x}{\sqrt{1+x^2}} dx = \sqrt{1+x^2} f(x) + A \ln |x + \sqrt{x^2+1}| + C$, then
 (A) $f(x) = \tan^{-1} x$, $A = -1$ (B) $f(x) = \tan^{-1} x$, $A = 1$
 (C) $f(x) = 2 \tan^{-1} x$, $A = -1$ (D) $f(x) = 2 \tan^{-1} x$, $A = 1$
16. $\int \frac{x + \sqrt{x+1}}{x+2} dx$ is equal to
 (A) $(x+1) - 2\sqrt{x+1} + 2 \ln|x+2| - 2 \tan^{-1} \sqrt{x+1} + C$
 (B) $(x+1) + 2\sqrt{x+2} - 2 \ln|x+2| - 2 \tan^{-1} \sqrt{x+2} + C$
 (C) $(x+1) + 2\sqrt{x+1} - 2 \ln|x+2| - 2 \tan^{-1} \sqrt{x+1} + C$
 (D) $(x+1) + 2\sqrt{x+2} - 2 \ln|x+1| + 2 \tan^{-1} \sqrt{x+2} + C$
17. The value of $\int \sqrt{\frac{1-\cos x}{\cos \alpha - \cos x}} dx$, where $0 < \alpha < x < \pi$, is equal to
 (A) $2 \ln \left(\cos \frac{\alpha}{2} - \cos \frac{x}{2} \right) + C$ (B) $\sqrt{2} \ln \left(\cos \frac{\alpha}{2} - \cos \frac{x}{2} \right) + C$
 (C) $2\sqrt{2} \ln \left(\cos \frac{\alpha}{2} - \cos \frac{x}{2} \right) + C$ (D) $-2 \sin^{-1} \left(\frac{\cos \frac{x}{2}}{\cos \frac{\alpha}{2}} \right) + C$
18. If $I = \int \frac{\sin x + \sin^3 x}{\cos 2x} dx = A \cos x + B \ln |f(x)| + C$, then
 (A) $A = \frac{1}{4}$, $B = \frac{-1}{\sqrt{2}}$, $f(x) = \frac{\sqrt{2} \cos x - 1}{\sqrt{2} \cos x + 1}$ (B) $A = -\frac{1}{2}$, $B = \frac{-3}{4\sqrt{2}}$, $f(x) = \frac{\sqrt{2} \cos x - 1}{\sqrt{2} \cos x + 1}$
 (C) $A = -\frac{1}{2}$, $B = \frac{3}{\sqrt{2}}$, $f(x) = \frac{\sqrt{2} \cos x + 1}{\sqrt{2} \cos x - 1}$ (D) $A = \frac{1}{2}$, $B = \frac{-3}{4\sqrt{2}}$, $f(x) = \frac{\sqrt{2} \cos x - 1}{\sqrt{2} \cos x + 1}$
19. The value of $\int \frac{1}{\cos^6 x + \sin^6 x} dx$ is equal to
 (A) $\tan^{-1} (\tan x + \cot x) + C$ (B) $-\tan^{-1} (\tan x + \cot x) + C$
 (C) $\tan^{-1} (\tan x - \cot x) + C$ (D) $-\tan^{-1} (\tan x - \cot x) + C$
20. Consider the following statements :
 S_1 : The antiderivative of every even function is an odd function.
 S_2 : Primitive of $\frac{3x^4 - 1}{(x^4 + x + 1)^2}$ w.r.t. x is $\frac{x}{x^4 + x + 1} + C$.
 S_3 : $\int \frac{1}{\sqrt{\sin^3 x \cos x}} dx = \frac{-2}{\sqrt{\tan x}} + C$.
 S_4 : The value of $\int \left(\sqrt{\frac{a+x}{a-x}} - \sqrt{\frac{a-x}{a+x}} \right) dx$ is equal to $-2 \sqrt{a^2 - x^2} + C$
 State, in order, whether S_1, S_2, S_3, S_4 are true or false
 (A) FTTT (B) TTTT (C) FFFF (D) TFTF
21. If $I_n = \int (\sin x + \cos x)^n dx$, and $I_n = \frac{1}{n} (\sin x + \cos x)^{n-1} (\sin x - \cos x) + \frac{2k}{n} I_{n-2}$ then $k =$
 (A) $(n+1)$ (B) $(n-1)$ (C) $(2n+1)$ (D) $(2n-1)$



PART-II: NUMERICAL VALUE QUESTIONS

INSTRUCTION :

- ❖ The answer to each question is **NUMERICAL VALUE** with two digit integer and decimal upto two digit.
- ❖ If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal placed.

* In each question C is arbitrary constant

1. If $f(x) = \int \frac{2\sin x - \sin 2x}{x^3} dx$, where $x \neq 0$, then $\lim_{x \rightarrow 0} f'(x)$ has the value
2. If $\int \sin^4 x \cos^4 x dx = \frac{1}{k} \left[ax - \sin 4x + \frac{1}{8} \sin 8x \right] + C$ then value of $\frac{k}{a}$ is
3. Let $f(x)$ be the primitive of $\frac{3x+2}{\sqrt{x-9}}$ w.r. to x . If $f(10) = 60$ then value of $f(11)$ is.
4. If $\int \frac{\sqrt{4+x^2}}{x^6} dx = \frac{(a+x^2)^{3/2} \cdot (x^2-b)}{120x^5} + C$ then $a^2 + b^2$ equals to :
5. If $\int \frac{\sqrt{x}}{\sqrt{a^3-x^3}} dx = k \sin^{-1} \left(\frac{x^{3/2}}{a^{3/2}} \right) + C$, then k equals to.
6. If $\int \frac{x dx}{\sqrt{1+x^2} \sqrt{(1+x^2)^3}} = k \sqrt{1+\sqrt{1+x^2}} + C$ then k^4 equals to :
7. If $\int e^{\sin x} \cdot \frac{x \cos^3 x - \sin x}{\cos^2 x} dx = e^{\sin x} f(x) + C$ such that $f(0) = -1$ then value of $f(4\pi)$ is
8. Let $g(x) = \int \frac{1+2\cos x}{(\cos x + 2)^2} dx$ and $g(0) = 0$ then value of $g\left(\frac{\pi}{2}\right)$ is.
9. If $f(x) = \sqrt{x-1}$; $g(x) = e^x$ and $\int f \circ g(x) dx = A f \circ g(x) + B \tan^{-1}(f \circ g(x)) + C$ then $A^3 + B^2$ equals
10. If $\int \frac{2 \sin 2\phi - \cos \phi}{6 - \cos^2 \phi - 4 \sin \phi} d\phi = p \ln |\sin^2 \phi - 4 \sin \phi + 5| + q \tan^{-1}(\sin \phi - r) + C$ then value of $\frac{p^2 + q^2}{r}$ is
11. If $\int \frac{(x-1)^2}{x^4 + x^2 + 1} dx = \frac{1}{\sqrt{a}} \tan^{-1} \left(\frac{x^2-1}{x\sqrt{3}} \right) - \frac{b}{\sqrt{a}} \tan^{-1} \left(\frac{2x^2+1}{\sqrt{3}} \right) + C$ then $a^2 + b^2$ equals to :
12. If $\int \frac{1+x \cos x}{x(1-x^2 e^{2\sin x})} dx = k \ln \sqrt{\frac{x^2 e^{2\sin x}}{1-x^2 e^{2\sin x}}} + C$ then k is equal to :
13. If $\int \frac{x^4+1}{x(x^2+1)^2} dx = A \ln |x| + \frac{B}{1+x^2} + C$, then $A + B$ equals to :
14. If $\int \frac{1}{1-\sin^4 x} dx = \frac{1}{a\sqrt{b}} \tan^{-1}(\sqrt{a} \tan x) + \frac{1}{b} \tan x + C$ then value of $a^3 + b^3$ is:



15. If $\int \frac{\cos^3 x + \cos^5 x}{\sin^2 x + \sin^4 x} dx = p \sin x - \frac{q}{\sin x} - r \tan^{-1}(\sin x) + C$ then value of $\frac{p+2r}{q}$ is
16. If $\int \frac{dx}{\sqrt{\sin^3 x \cos^5 x}} = a \sqrt{\cot x} + b \sqrt{\tan^3 x} + C$, where C is an arbitrary constant of integration, then the values of $a^2 + 10b$ equals to :

PART - III : ONE OR MORE THAN ONE OPTIONS CORRECT TYPE

* In each question C is arbitrary constant

1. The value of $\int 2^{mx} \cdot 3^{nx} dx$ (when $m, n \in \mathbb{N}$) is equal to :
- (A) $\frac{2^{mx} + 3^{nx}}{m \ln 2 + n \ln 3} + C$ (B) $\frac{e^{(m \ln 2 + n \ln 3)x}}{m \ln 2 + n \ln 3} + C$ (C) $\frac{2^{mx} \cdot 3^{nx}}{\ln(2^m \cdot 3^n)} + C$ (D) $\frac{(mn) \cdot 2^x \cdot 3^x}{m \ln 2 + n \ln 3} + C$
2. If $f\left(\frac{1-x}{1+x}\right) = x$ and $g(x) = \int f(x) dx$ then
- (A) $g(x)$ is continuous in domain (B) $g(x)$ is discontinuous at two points in its domain
- (C) $\lim_{x \rightarrow \infty} g'(x) = -1$ (D) $\int g(x) dx = -\frac{x^2}{2} + (2x+1) \lambda n\left(\frac{1+x}{e}\right) + C$
3. If $\int \tan^5 x dx = A \tan^4 x - \frac{1}{2} \tan^2 x + B \ln|\sec x| + C$ then
- (A) $A = \frac{1}{4}$ (B) $A = \frac{1}{2}$ (C) $B = 1$ (D) $B = -1$
4. The value of $\int \{1 + 2 \tan x (\tan x + \sec x)\}^{1/2} dx$ is equal to
- (A) $\ln |\sec x (\sec x - \tan x)| + C$ (B) $\ln |\operatorname{cosec} x (\sec x + \tan x)| + C$
- (C) $\ln |\sec x (\sec x + \tan x)| + C$ (D) $-\ln |\cos x (\sec x - \tan x)| + C$
5. The value of $\int \frac{\ln\left(\frac{x-1}{x+1}\right)}{x^2 - 1} dx$ is equal to
- (A) $\frac{1}{2} \ln^2 \frac{x-1}{x+1} + C$ (B) $\frac{1}{4} \ln^2 \frac{x-1}{x+1} + C$ (C) $\frac{1}{2} \ln^2 \frac{x+1}{x-1} + C$ (D) $\frac{1}{4} \ln^2 \frac{x+1}{x-1} + C$
6. The value of $\int \frac{\ln(\tan x)}{\sin x \cos x} dx$ is equal to
- (A) $\frac{1}{2} \ln^2 (\cot x) + C$ (B) $\frac{1}{2} \ln^2 (\sec x) + C$
- (C) $\frac{1}{2} \ln^2 (\sin x \sec x) + C$ (D) $\frac{1}{2} \ln^2 (\cos x \operatorname{cosec} x) + C$
7. The value of $\int \frac{\cos^3 x}{\sin^2 x + \sin x} dx$ is equal to :
- (A) $\ln |\sin x| + \sin x + C$ (B) $\ln |\sin x| - \sin x + C$
- (C) $-\ln |\operatorname{cosec} x| - \sin x + C$ (D) $-\ln |\sin x| + \sin x + C$



8. If $\int \frac{(x-1) dx}{x^2 \sqrt{2x^2 - 2x + 1}} = \frac{\sqrt{f(x)}}{g(x)} + C$, where $f(x)$ is a quadratic expression and $g(x)$ is a monic linear expression.
- (A) $f(x) = 2x^2 - 2x + 1$ (B) $g(x) = x + 1$ (C) $g(x) = x$ (D) $f(x) = 2x^2 - 2x$
9. If $\int e^{3x} \cos 4x dx = e^{3x} (A \sin 4x + B \cos 4x) + C$ then :
- (A) $4A = 3B$ (B) $2A = 3B$ (C) $3A = 4B$ (D) $4A + 3B = 1$
10. $I = \int \frac{\sin^{-1} \sqrt{x} - \cos^{-1} \sqrt{x}}{\sin^{-1} \sqrt{x} + \cos^{-1} \sqrt{x}} dx$ equals to
- (A) $-x + \frac{2}{\pi} (2x - 1) \sin^{-1} \sqrt{x} + \frac{2}{\pi} \sqrt{x - x^2} + C$
- (B) $x - \frac{4x}{\pi} \cos^{-1} \sqrt{x} - \frac{2}{\pi} \sin^{-1} \sqrt{x} + \frac{2}{\pi} \sqrt{x} \sqrt{1-x} + C$
- (C) $-x + \frac{2}{\pi} (2x + 1) \cos^{-1} \sqrt{x} + \frac{2}{\pi} \sqrt{x} \sqrt{1-x} + C$
- (D) $x - \frac{4x}{\pi} \sin^{-1} \sqrt{x} + C$
11. If $\int \frac{x^2 - x + 1}{(1 + x^2)^{3/2}} e^x dx = e^x f(x) + C$ then
- (A) $f(x)$ is an even function (B) $f(x)$ is a bounded function
- (C) Range of $f(x)$ is $(0, 1]$ (D) $f(x)$ has two points of extrema.
12. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B \ln |9e^{2x} - 4| + C$, then
- (A) $A + 18B = 16$ (B) $18B - A = 19$ (C) $A - 18B = 17$ (D) $A + 18B = 32$
13. The value of $\int \frac{x^2 + \cos^2 x}{1 + x^2} \operatorname{cosec}^2 x dx$ is equal to:
- (A) $\cot x - \cot^{-1} x + C$ (B) $C - \cot x + \cot^{-1} x$
- (C) $-\tan^{-1} x - \frac{\operatorname{cosec} x}{\sec x} + C$ (D) $\frac{1}{\tan^{-1} x} - \cot x + C$
14. The value of $\int \frac{dx}{\sqrt{x - x^2}}; \left(x > \frac{1}{2}\right)$ is equal to
- (A) $2 \sin^{-1} \sqrt{x} + C$ (B) $\sin^{-1} (2x - 1) + C$
- (C) $C - 2 \cos^{-1} (2x - 1)$ (D) $\cos^{-1} 2\sqrt{x - x^2} + C$
15. $\int \frac{x^3 - 1}{x^3 + x} dx$ is equal to
- (A) $x - \ln |x| + \ln (x^2 + 1) - \tan^{-1} x + C$ (B) $x - \ln |x| + \frac{1}{2} \ln (x^2 + 1) - \tan^{-1} x + C$
- (C) $x + \ln |x| + \frac{1}{2} \ln (x^2 + 1) + \tan^{-1} x + C$ (D) $x + \ln \sqrt{\frac{x^2 + 1}{x^2}} + \cot^{-1} x + C$



16. The value of $2 \int \sin x \cdot \operatorname{cosec} 4x \, dx$ is equal to

- (A) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$ (B) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1+\sqrt{2}\sin x}{1-\sqrt{2}\sin x} \right| - \frac{1}{2} \ln \left| \frac{1+\sin x}{\cos x} \right| + C$
 (C) $\frac{1}{2\sqrt{2}} \ln \left| \frac{1-\sqrt{2}\sin x}{1+\sqrt{2}\sin x} \right| - \frac{1}{4} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$ (D) $-\frac{1}{2\sqrt{2}} \ln \left| \frac{1-\sqrt{2}\sin x}{1+\sqrt{2}\sin x} \right| + \frac{1}{4} \ln \left| \frac{1-\sin x}{1+\sin x} \right| + C$

17. If $\int \frac{3\cot 3x - \cot x}{\tan x - 3\tan 3x} \, dx = p f(x) + q g(x) + C$, then which of the following may be correct?

- (A) $p = 1; q = \frac{1}{\sqrt{3}}; f(x) = x; g(x) = \ln \left| \frac{\sqrt{3} - \tan x}{\sqrt{3} + \tan x} \right|$
 (B) $p = 1; q = -\frac{1}{\sqrt{3}}; f(x) = x; g(x) = \ln \left| \frac{\sqrt{3} - \tan x}{\sqrt{3} + \tan x} \right|$
 (C) $p = 1; q = -\frac{2}{\sqrt{3}}; f(x) = x; g(x) = \ln \left| \frac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x} \right|$
 (D) $p = 1; q = -\frac{1}{\sqrt{3}}; f(x) = x; g(x) = \ln \left| \frac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x} \right|$

18. If $\int \frac{dx}{5 + 4\cos x} = P \tan^{-1} \left(m \tan \frac{x}{2} \right) + C$ then :

- (A) $P = 2/3$ (B) $m = 1/3$ (C) $P = 1/3$ (D) $m = 2/3$

19. The value of $\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} \, dx$ is equal to:

- (A) $\cot^{-1}(\cot^2 x) + C$ (B) $-\cot^{-1}(\tan^2 x) + C$ (C) $\tan^{-1}(\tan^2 x) + C$ (D) $-\tan^{-1}(\cos 2x) + C$

PART - IV : COMPREHENSION

Comprehension # 1 (Q.No. 1 to 3)

Let $I_{n,m} = \int \sin^n x \cos^m x \, dx$. Then we can relate $I_{n,m}$ with each of the following

- (i) $I_{n-2,m}$ (ii) $I_{n+2,m}$ (iii) $I_{n,m-2}$
 (iv) $I_{n,m+2}$ (v) $I_{n-2,m+2}$ (vi) $I_{n+2,m-2}$

Suppose we want to establish a relation between $I_{n,m}$ and $I_{n,m-2}$, then we set

$$P(x) = \sin^{n+1} x \cos^{m-1} x \quad \dots\dots\dots (1)$$

In $I_{n,m}$ and $I_{n,m-2}$ the exponent of $\cos x$ is m and $m-2$ respectively, the minimum of the two is $m-2$, adding 1 to the minimum we get $m-2+1 = m-1$. Now choose the exponent $m-1$ of $\cos x$ in $P(x)$. Similarly choose the exponent of $\sin x$ for $P(x)$

Now differentiating both sides of (1), we get

$$\begin{aligned} P'(x) &= (n+1) \sin^n x \cos^m x - (m-1) \sin^{n+2} x \cos^{m-2} x \\ &= (n+1) \sin^n x \cos^m x - (m-1) \sin^n x (1 - \cos^2 x) \cos^{m-2} x \\ &= (n+1) \sin^n x \cos^m x - (m-1) \sin^n x \cos^{m-2} x + (m-1) \sin^n x \cos^m x \\ &= (n+m) \sin^n x \cos^m x - (m-1) \sin^n x \cos^{m-2} x \end{aligned}$$

Now integrating both sides, we get

$$\sin^{n+1} x \cos^{m-1} x = (n+m) I_{n,m} - (m-1) I_{n,m-2}$$

Similarly we can establish the other relations.





1. The relation between $I_{4,2}$ and $I_{2,2}$ is

(A) $I_{4,2} = \frac{1}{6} (-\sin^3 x \cos^3 x + 3I_{2,2})$

(B) $I_{4,2} = \frac{1}{6} (\sin^3 x \cos^3 x + 3I_{2,2})$

(C) $I_{4,2} = \frac{1}{6} (\sin^3 x \cos^3 x - 3I_{2,2})$

(D) $I_{4,2} = \frac{1}{6} (-\sin^3 x \cos^3 x + 2I_{2,2})$

2. The relation between $I_{4,2}$ and $I_{6,2}$ is

(A) $I_{4,2} = \frac{1}{5} (\sin^5 x \cos^3 x + 8I_{6,2})$

(B) $I_{4,2} = \frac{1}{5} (-\sin^5 x \cos^3 x + 8I_{6,2})$

(C) $I_{4,2} = \frac{1}{5} (\sin^5 x \cos^3 x - 8I_{6,2})$

(D) $I_{4,2} = \frac{1}{5} (\sin^5 x \cos^3 x + 8I_{6,2})$

3. The relation between $I_{4,2}$ and $I_{4,4}$ is

(A) $I_{4,2} = \frac{1}{3} (\sin^5 x \cos^3 x + 8I_{4,4})$

(B) $I_{4,2} = \frac{1}{3} (-\sin^5 x \cos^3 x + 8I_{4,4})$

(C) $I_{4,2} = \frac{1}{3} (\sin^5 x \cos^3 x - 8I_{4,4})$

(D) $I_{4,2} = \frac{1}{3} (\sin^5 x \cos^3 x + 6I_{4,4})$

Comprehension # 2 (Q. No. 4 to 6)

It is known that

$$\sqrt{\tan x} + \sqrt{\cot x} = \begin{cases} \frac{\sqrt{\sin x}}{\sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x}} & \text{if } 0 < x < \frac{\pi}{2} \\ \frac{\sqrt{-\sin x}}{\sqrt{-\cos x}} + \frac{\sqrt{-\cos x}}{\sqrt{-\sin x}} & \text{if } \pi < x < \frac{3\pi}{2} \end{cases},$$

$$\frac{d}{dx} (\sqrt{\tan x} - \sqrt{\cot x}) = \frac{1}{2} (\sqrt{\tan x} + \sqrt{\cot x}) (\tan x + \cot x), \forall x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$$

$$\text{and } \frac{d}{dx} (\sqrt{\tan x} + \sqrt{\cot x}) = \frac{1}{2} (\sqrt{\tan x} - \sqrt{\cot x}) (\tan x + \cot x), \forall x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right).$$

4. Value of integral $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$ is

(A) $\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C$

(B) $\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{2}} \right) + C$

(C) $-\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} - \sqrt{\cot x}}{\sqrt{2}} \right) + C$

(D) $-\sqrt{2} \tan^{-1} \left(\frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{2}} \right) + C$

5. Value of the integral $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(0, \frac{\pi}{2}\right)$, is

(A) $\sqrt{2} \sin^{-1} (\cos x - \sin x) + C$

(B) $\sqrt{2} \sin^{-1} (\sin x - \cos x) + C$

(C) $\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$

(D) $-\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$

6. Value of the integral $I = \int (\sqrt{\tan x} + \sqrt{\cot x}) dx$, where $x \in \left(\pi, \frac{3\pi}{2}\right)$, is

(A) $\sqrt{2} \sin^{-1} (\cos x - \sin x) + C$

(B) $\sqrt{2} \sin^{-1} (\sin x - \cos x) + C$

(C) $\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$

(D) $-\sqrt{2} \sin^{-1} (\sin x + \cos x) + C$





Exercise-3

✎ Marked questions are recommended for Revision.

* Marked Questions may have more than one correct option.

PART - I : JEE (ADVANCED) / IIT-JEE PROBLEMS (PREVIOUS YEARS)

- Let $f(x) = \int e^x (x-1)(x-2) dx$ then f decreases in the interval: [IIT-JEE 2000, Scr, (1, 0), 35]
 (A) $(-\infty, 2)$ (B) $(-2, -1)$ (C) $(1, 2)$ (D) $(2, +\infty)$
- Evaluate, $\int \sin^{-1} \left(\frac{2x+2}{\sqrt{4x^2+8x+13}} \right) dx$. [IIT-JEE 2001, Main, (5, 0), 100]
- ✎ For any natural number m , evaluate,
 $\int (x^{3m} + x^{2m} + x^m) (2x^{2m} + 3x^m + 6)^{1/m} dx, x > 0$. [IIT-JEE 2002, Main, (5, 0), 60]
- $\int \frac{x^2-1}{x^3\sqrt{2x^4-2x^2+1}} dx$ is equal to [IIT-JEE 2006, (3, -1), 184]
 (A) $\frac{\sqrt{2x^4-2x^2+1}}{x^2} + C$ (B) $\frac{\sqrt{2x^4-2x^2+1}}{x^3} + C$ (C) $\frac{\sqrt{2x^4-2x^2+1}}{x} + C$ (D) $\frac{\sqrt{2x^4-2x^2+1}}{2x^2} + C$
- ✎ Let $f(x) = \frac{x}{(1+x^n)^{1/n}}$ for $n \geq 2$ and $g(x) = \underbrace{(f \circ f \circ \dots \circ f)}_{f \text{ occurs } n \text{ times}}(x)$. Then $\int x^{n-2} g(x) dx$ equals [IIT-JEE 2007, Paper-2, (3, -1), 81]
 (A) $\frac{1}{n(n-1)} (1+nx^n)^{1-\frac{1}{n}} + K$ (B) $\frac{1}{(n-1)} (1+nx^n)^{1-\frac{1}{n}} + K$
 (C) $\frac{1}{n(n+1)} (1+nx^n)^{1+\frac{1}{n}} + K$ (D) $\frac{1}{(n+1)} (1+nx^n)^{1+\frac{1}{n}} + K$
- Let $F(x)$ be an indefinite integral of $\sin^2 x$. [IIT-JEE 2007, Paper-1, (3, -1), 81]
 STATEMENT-1 : The function $F(x)$ satisfies $F(x + \pi) = F(x)$ for all real x .
because
 STATEMENT-2 : $\sin^2(x + \pi) = \sin^2 x$ for all real x .
 (A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1
 (B) Statement-1 is True, Statement-2 is True ; Statement-2 is **NOT** a correct explanation for Statement 1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
- Let $I = \int \frac{e^x}{e^{4x} + e^{2x} + 1} dx$, $J = \int \frac{e^{-x}}{e^{-4x} + e^{-2x} + 1} dx$. Then, for an arbitrary constant C , the value of $J - I$ is equal to : [IIT-JEE 2008, Paper-2, (3, -1), 81]
 (A) $\frac{1}{2} \ln \left| \frac{e^{4x} - e^{2x} + 1}{e^{4x} + e^{2x} + 1} \right| + C$ (B) $\frac{1}{2} \ln \left| \frac{e^{2x} + e^x + 1}{e^{2x} - e^x + 1} \right| + C$
 (C) $\frac{1}{2} \ln \left| \frac{e^{2x} - e^x + 1}{e^{2x} + e^x + 1} \right| + C$ (D) $\frac{1}{2} \ln \left| \frac{e^{4x} + e^{2x} + 1}{e^{4x} - e^{2x} + 1} \right| + C$



8. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K)
- (A) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$ [IIT-JEE 2012, Paper-1, (3, -1), 70]
- (B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
- (C) $\frac{-1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$
- (D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

PART - II : JEE (MAIN) / AIEEE PROBLEMS (PREVIOUS YEARS)

1. If the integral $\int \frac{5 \tan x}{\tan x - 2} dx = x + a \ln |\sin x - 2 \cos x| + k$, then a is equal to : [AIEEE-2012, (4, -1)/120]
- (1) -1 (2) -2 (3) 1 (4) 2
2. If $\int f(x) dx = \psi(x)$, then $\int x^5 f(x^3) dx$ is equal to [AIEEE - 2013, (4, -1), 360]
- (1) $\frac{1}{3} [x^3 \psi(x^3) - \int x^2 \psi(x^3) dx] + C$ (2) $\frac{1}{3} x^3 \psi(x^3) - 3 \int x^3 \psi(x^3) dx + C$
- (3) $\frac{1}{3} x^3 \psi(x^3) - \int x^2 \psi(x^3) dx + C$ (4) $\frac{1}{3} [x^3 \psi(x^3) - \int x^3 \psi(x^3) dx] + C$
3. The integral $\int \left(1 + x - \frac{1}{x} \right) e^{x + \frac{1}{x}} dx$ is equal to : [JEE(Main) 2014, (4, -1), 120]
- (1) $(x+1) e^{x + \frac{1}{x}} + c$ (2) $-x e^{x + \frac{1}{x}} + c$ (3) $(x-1) e^{x + \frac{1}{x}} + c$ (4) $x e^{x + \frac{1}{x}} + c$
4. The integral $\int \frac{dx}{x^2 (x^4 + 1)^{3/4}}$ equals [JEE(Main) 2015, (4, -1), 120]
- (1) $\left(\frac{x^4 + 1}{x^4} \right)^{1/4} + c$ (2) $(x^4 + 1)^{1/4} + c$ (3) $-(x^4 + 1)^{1/4} + c$ (4) $-\left(\frac{x^4 + 1}{x^4} \right)^{1/4} + c$
5. The integral $\int \frac{2x^{12} + 5x^9}{(x^5 + x^3 + 1)^3} dx$ is equal to [JEE(Main) 2016, (4, -1), 120]
- (1) $\frac{x^{10}}{2(x^5 + x^3 + 1)^2} + C$ (2) $\frac{x^5}{2(x^5 + x^3 + 1)^2} + C$ (3) $\frac{-x^{10}}{2(x^5 + x^3 + 1)^2} + C$ (4) $\frac{-x^5}{(x^5 + x^3 + 1)^2} + C$
- where C is an arbitrary constant
6. Let $I_n = \int \tan^n x dx$, ($n > 1$). If $I_4 + I_6 = a \tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to [JEE(Main) 2017, (4, -1), 120]
- (1) $\left(-\frac{1}{5}, 1\right)$ (2) $\left(\frac{1}{5}, 0\right)$ (3) $\left(\frac{1}{5}, -1\right)$ (4) $\left(-\frac{1}{5}, 0\right)$
7. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is equal to : [JEE(Main) 2018, (4, -1), 120]
- (1) $\frac{1}{1 + \cot^3 x} + C$ (2) $\frac{-1}{1 + \cot^3 x} + C$ (3) $\frac{1}{3(1 + \tan^3 x)} + C$ (4) $\frac{-1}{3(1 + \tan^3 x)} + C$
- (where C is a constant of integration)



8. Let $n \geq 2$ be a natural number and $0 < \theta < \pi/2$. Then $\int \frac{(\sin^n \theta - \sin \theta)^{\frac{1}{n}} \cos \theta}{\sin^{n+1} \theta} d\theta$ is equal to :
(where C is a constant of integration) [JEE(Main) 2019, Online (10-01-19), P-1 (4, -1), 120]
- (1) $\frac{n}{n^2-1} \left(1 - \frac{1}{\sin^{n+1} \theta}\right)^{\frac{n+1}{n}} + C$ (2) $\frac{n}{n^2-1} \left(1 - \frac{1}{\sin^{n-1} \theta}\right)^{\frac{n+1}{n}} + C$
(3) $\frac{n}{n^2+1} \left(1 - \frac{1}{\sin^{n-1} \theta}\right)^{\frac{n+1}{n}} + C$ (4) $\frac{n}{n^2-1} \left(1 + \frac{1}{\sin^{n-1} \theta}\right)^{\frac{n+1}{n}} + C$
9. The integral $\int \cos(\log_e x) dx$ is equal to : (where C is a constant of integration) [JEE(Main) 2019, Online (12-01-19), P-1 (4, -1), 120]
- (1) $x[\cos(\log_e x) - \sin(\log_e x)] + C$ (2) $\frac{x}{2}[\sin(\log_e x) - \cos(\log_e x)] + C$
(3) $x[\cos(\log_e x) + \sin(\log_e x)] + C$ (4) $\frac{x}{2}[\cos(\log_e x) + \sin(\log_e x)] + C$
10. $\int \frac{\sin \frac{5x}{2}}{\sin \frac{x}{2}} dx$ is equal to : (where c is a constant of integration) [JEE(Main) 2019, Online (08-04-19), P-1 (4, -1), 120]
- (1) $x + 2 \sin x + 2 \sin 2x + c$ (2) $2x + \sin x + \sin 2x + c$
(3) $2x + \sin x + 2 \sin 2x + c$ (4) $x + 2 \sin x + \sin 2x + c$
11. If $\int e^{\sec x} (\sec x \tan x f(x) + (\sec x \tan x + \sec^2 x)) dx = e^{\sec x} f(x) + C$, then a possible choice of $f(x)$ is : [JEE(Main) 2019, Online (09-04-19), P-2 (4, -1), 120]
- (1) $\sec x + x \tan x - \frac{1}{2}$ (2) $\sec x + \tan x + \frac{1}{2}$ (3) $x \sec x + \tan x + \frac{1}{2}$ (4) $\sec x - \tan x - \frac{1}{2}$
12. If $\int \frac{dx}{(x^2 - 2x + 10)^2} = A \left(\tan^{-1} \left(\frac{x-1}{3} \right) + \frac{f(x)}{x^2 - 2x + 10} \right) + C$
Where C is a constant of integration, then [JEE(Main) 2019, Online (10-04-19), P-1 (4, -1), 120]
- (1) $A = \frac{1}{27}$ and $f(x) = 9(x-1)$ (2) $A = \frac{1}{54}$ and $f(x) = 9(x-1)^2$
(3) $A = \frac{1}{81}$ and $f(x) = 3(x-1)$ (4) $A = \frac{1}{54}$ and $f(x) = 3(x-1)$
13. If $\int \frac{d\theta}{\cos^2 \theta (\tan 2\theta + \sec 2\theta)} = \lambda \tan \theta + 2 \log_e |f(\theta)| + C$ where C is a constant of integration, then the ordered pair $(\lambda, f(\theta))$ is equal to : [JEE(Main) 2020, Online (09-01-20), P-2 (4, -1), 120]
- (1) $(1, 1 + \tan \theta)$ (2) $(-1, 1 - \tan \theta)$ (3) $(-1, 1 + \tan \theta)$ (4) $(1, 1 - \tan \theta)$



Answers

EXERCISE - 1

PART-I

Section (A) :

- A-1.** (i) $\frac{(2x+3)^6}{12} + C$ (ii) $-\frac{\cos 2x}{2} + C$
 (iii) $\frac{\tan(4x+5)}{4} + C$ (iv) $\frac{1}{3} \ln |\sec(3x+2) + \tan(3x+2)| + C$
 (v) $\frac{1}{2} \ln |\sec(2x+1)| + C$ (vi) $\frac{2^{3x+4}}{3 \ln 2} + C$
 (vii) $\frac{1}{2} \ln |2x+1| + C$ (viii) $\frac{e^{4x+5}}{4} + C$

- A-2.** (i) $\frac{x}{2} - \frac{1}{4} \sin 2x + C$ (ii) $\frac{\sin 3x}{12} + \frac{3}{4} \sin x + C$
 (iii) $-\frac{1}{10} \cos 5x + \frac{1}{2} \cos x + C$ (iv) $\cos x - \frac{1}{2} \cos 2x - \frac{1}{3} \cos 3x + C$
 (v) $\frac{2}{3} ((x+3)^{3/2} + (x+2)^{3/2}) + C$

Section (B) :

- B-1.** (i) $-\frac{1}{2} \cos x^2 + C$ (ii) $\frac{1}{2} \ln |x^2 + 1| + C$
 (iii) $\frac{1}{2} (\tan x)^2 + C$ or $\frac{\sec^2 x}{2} + C$ (iv) $\ln |e^x + x| + C$
 (v) $\ln |x + \cos x| + C$ (vi) $\frac{1}{2} \ln |e^{2x} - 2| + C$
 (vii) $\frac{1}{2} \ln |x^2 + \sin 2x + 2x| + C$ (viii) $\ln |\ln(\sec x + \tan x)| + C$
 (ix) $\frac{2}{3} (x+2)^{3/2} - 4(x+2)^{1/2} + C$ (x) $\frac{1}{2} (e^{2x} - e^{-2x}) + 2x + C$
 (xi) $\frac{1}{3} e^{3x} + e^{2x} + e^x + C$ (xii) $-\frac{1}{5} \ln \left| 1 + \frac{1}{x^5} \right| + C$
 (xiii) $-\frac{1}{4} \left(1 + \frac{1}{x^5} \right)^{4/5} + C$ (xiv) $\frac{(x^2 - 8)^{3/2}}{24 x^3} + C$

B-2. $2\sqrt{(x^2+2)} + C$

B-3. (i) $\ln \left(\frac{\sin x}{x} \right) + C$ (ii) $\ln \left(\frac{\ln(x+1)}{x} \right) + C$



**Section (C) :**

- C-1.** (i) $\frac{x^2}{2} \ln x - \frac{x^2}{4} + C$ (ii) $\frac{x^2}{4} - \frac{x}{4} \sin 2x - \frac{1}{8} \cos 2x + C$
 (iii) $\frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{1}{2} \tan^{-1} x + C$ (iv) $x (\ln x - 1) + C$
 (v) $\frac{\sec x \tan x}{2} + \frac{1}{2} \ln |\sec x + \tan x| + C$ (vi) $(x^2 - 1) e^{x^2} + C$
 (vii) $x \sin^{-1} \sqrt{x} + \frac{\sqrt{x} \sqrt{1-x}}{2} - \frac{1}{2} \sin^{-1} \sqrt{x} + C$ (viii) $x \tan^{-1} x - \frac{1}{2} \ln(1+x^2) - \frac{(\tan^{-1} x)^2}{2} + C$
 (ix) $\frac{e^x}{2} (\sin x - \cos x) + C$ (x) $e^x \tan x + C$

C-2. $y = x \left[\ln(\ln x) - \frac{1}{\ln x} \right] + 2e$

Section (D) :

- D-1.** (i) $\frac{1}{2} \tan^{-1} \frac{x}{2} + C$ (ii) $\frac{1}{\sqrt{5}} \tan^{-1} \frac{x}{\sqrt{5}} + C$
 (iii) $\frac{1}{2} \tan^{-1} \left(\frac{(x+1)}{2} \right) + C$ (iv) $\ln |x^2 + 3x + 4| - \frac{4}{\sqrt{7}} \tan^{-1} \frac{2x+3}{\sqrt{7}} + C$
 (v) $x - \arctan x + \ln \frac{\sqrt{1+x^2}}{x} + C$ (vi) $\ln |x + \sqrt{x^2 - 4}| + C$
 (vii) $\frac{x}{2} \sqrt{x^2 + 4} + 2 \ln |x + \sqrt{x^2 + 4}| + C$ (viii) $\frac{x+1}{2} \sqrt{x^2 + 2x + 5} + 2 \ln |x+1 + \sqrt{x^2 + 2x + 5}| + C$
 (ix) $-\frac{(1-x-x^2)^{3/2}}{3} - \frac{3}{8} (2x+1) \sqrt{1-x-x^2} - \frac{15}{16} \sin^{-1} \left(\frac{2x+1}{\sqrt{5}} \right) + C$
 (x) $\frac{2}{15} (a^3 + x^3)^{5/2} - \frac{2a^3}{9} (a^3 + x^3)^{3/2} + C$
- D-2.** (i) $\ln \left| \frac{x+1}{x+2} \right| + C$ (ii) $\frac{1}{10} \ln |x+3| - \frac{1}{20} \ln |x^2+1| + \frac{3}{10} \tan^{-1} x + C$
 (iii) $4 \ln |x+1| + \frac{1}{(x+1)} - 4 \ln |x+2| + C$ (iv) $\frac{1}{2} \ln |x+1| - \ln |x+2| + \frac{1}{2} \ln |x+3| + C$
- D-3.** (i) $\frac{1}{2\sqrt{3}} \tan^{-1} \left(\frac{x^2-1}{\sqrt{3}x} \right) - \frac{1}{4} \ln \left| \frac{x+\frac{1}{x}-1}{x+\frac{1}{x}+1} \right| + C$ (ii) $\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{x^2-1}{\sqrt{2}x} \right) + C$
 (iii) $-\frac{1}{2\sqrt{3}} \ln \left| \frac{x+\frac{1}{x}-\sqrt{3}}{x+\frac{1}{x}+\sqrt{3}} \right| + C$
- D-4.** (i) $\ln \left| \frac{\sqrt{x+2}-1}{\sqrt{x+2}+1} \right| + C$ (ii) $\frac{1}{4\sqrt{3}} \ln \left| \frac{t-\sqrt{3}}{t+\sqrt{3}} \right| - \frac{1}{2} \tan^{-1}(t) + C$, where $t = \sqrt{x+1}$
 (iii) $-\frac{1}{\sqrt{3}} \ln \left| \left(t - \frac{1}{3} \right) + \sqrt{\left(t - \frac{1}{3} \right)^2 + \frac{2}{9}} \right| + C$, where $t = \frac{1}{x+1}$ (iv) $-\tan^{-1} \sqrt{\frac{x^2+2}{x^2}} + C$



D-5. (i) $\frac{1}{2} \ln \left| \left(x + \frac{1}{2} \right) + \sqrt{x^2 + x} \right| + \sqrt{x^2 + x} + C$

(ii) $\sqrt{x^2 - 1} - \ln |x + \sqrt{x^2 - 1}| + C$

(iii) $\frac{1}{2} \sin^{-1} x - \frac{x}{2} \sqrt{1 - x^2} - \sqrt{1 - x^2} + C$

Section (E) :

E-1. (i) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{\tan x / 2}{\sqrt{3}} \right) + C$

(ii) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\sqrt{3} \tan \frac{x}{2} \right) + C$

(iii) $\frac{10}{13} x - \frac{2}{13} \ln |3 \cos x + 2 \sin x| + C$

(iv) $\ln \left| 1 + \tan \frac{x}{2} \right| + C$

(v) $\frac{1}{\sqrt{6}} \tan^{-1} \left(\frac{\sqrt{3} \tan x}{\sqrt{2}} \right) + C$

(vi) $\ln |1 - \cot x| + C$

(vii) $\tan x + \frac{1}{4} \sin 2x - \frac{3x}{2} + C$

E-2. (i) $\frac{1}{40} \ln \left(\frac{4(\sin x - \cos x) + 5}{4(\sin x + \cos x) - 5} \right) + C$

(ii) $\sin^{-1} \left(\frac{\sin x + \cos x}{3} \right) + C$

E-3. $A = \frac{1}{9}$, $B = \frac{1}{5}$

PART-II

Section (A) :

A-1. (D) **A-2.** (A) **A-3.** (A) **A-4.** (B) **A-5.** (A) **A-6.** (B)

Section (B) :

B-1. (B) **B-2.** (C) **B-3.** (A) **B-4.** (D) **B-5.** (D) **B-6.** (C) **B-7.** (A)
B-8. (C)

Section (C) :

C-1. (C) **C-2.** (A) **C-3.** (C) **C-4.** (A) **C-5.** (C)

Section (D) :

D-1. (B) **D-2.** (D) **D-3.** (A) **D-4.** (A) **D-5.** (C)

Section (E) :

E-1. (B) **E-2.** (C) **E-3.** (C) **E-4.** (B) **E-5.** (A) **E-6.** (A) **E-7.** (A)

Section (F) :

F-1. (B) **F-2.** (B)

PART-III

- (A) → (p), (B) → (p), (C) → (r), (D) → (s)
- (A) → (s); (B) → (q); (C) → (r)

EXERCISE - 2

PART-I

- | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|
| 1. (A) | 2. (B) | 3. (B) | 4. (C) | 5. (C) | 6. (A) | 7. (A) |
| 8. (A) | 9. (A) | 10. (C) | 11. (A) | 12. (D) | 13. (A) | 14. (A) |
| 15. (A) | 16. (C) | 17. (D) | 18. (D) | 19. (C) | 20. (A) | 21. (B) |

**PART-II**

1. 01.00 2. 42.66 or 42.67 3. 87.68 4. 52.00 5. 00.66 or 00.67
 6. 16.00 7. 11.56 or 11.57 8. 00.50 9. 12.00 10. 26.50
 11. 13.00 12. 01.00 13. 02.00 14. 16.00 15. 06.50
 16. 10.66 or 1067

PART-III

1. (BC) 2. (AC) 3. (AC) 4. (CD) 5. (BD) 6. (ACD) 7. (BC)
 8. (AC) 9. (CD) 10. (AB) 11. (ABC) 12. (AB) 13. (BC) 14. (ABD)
 15. (BD) 16. (ABD) 17. (AD) 18. (AB) 19. (ABCD)

PART-IV

1. (A) 2. (A) 3. (B) 4. (A) 5. (B) 6. (A)

EXERCISE - 3**PART-I**

1. (C) 2. $(x+1)\tan^{-1}\left(\frac{2x+2}{3}\right) - \frac{3}{4} \ln(4x^2+8x+13) + C$
 3. $\frac{(2x^{3m}+3x^{2m}+6x^m)^{\frac{m+1}{m}}}{6(m+1)} + C$ 4. (D) 5. (A) 6. (D) 7. (C)
 8. (C)

PART-II

1. (4) 2. (3) 3. (4) 4. (4) 5. (1) 6. (2) 7. (4)
 8. (2) 9. (4) 10. (4) 11. (2) 12. (4) 13. (3)



High Level Problems (HLP)

1. Evaluate : $\int \frac{\sin^8 x - \cos^8 x}{1 - 2\sin^2 x \cos^2 x} dx$
2. Evaluate : $\int \frac{\cos 5x + \cos 4x}{1 - 2\cos 3x} dx$
3. Evaluate : $\int \sqrt{x + \sqrt{x^2 + 2}} dx$
4. Evaluate : $\int \frac{dx}{(x^3 + 3x^2 + 3x + 1) \sqrt{x^2 + 2x - 3}}$
5. Evaluate : $\int \frac{(\cos 2x - 3)}{\cos^4 x \sqrt{4 - \cot^2 x}} dx$
6. Evaluate : $\left[\frac{\sqrt{x^2 + 1} \{ \ln(x^2 + 1) - 2 \ln x \}}{x^4} \right] dx$
7. Evaluate : $\int \frac{x}{(7x - 10 - x^2)^{3/2}} dx$
8. If $\int \frac{x \cos \alpha + 1}{(x^2 + 2x \cos \alpha + 1)^{3/2}} dx = \frac{f(x)}{\sqrt{g(x)}} + C$ then find $f(x)$ and $g(x)$.
9. Evaluate : $\cos x \cdot e^x \cdot x^2 dx$
10. Evaluate : $\int e^x \left(\frac{x^3 - x + 2}{(x^2 + 1)^2} \right) dx$
11. Evaluate : $\int \frac{x^2}{(x \sin x + \cos x)^2} dx$
12. Evaluate : $\int \sin 4x \cdot e^{\tan^2 x} dx$
13. Evaluate : $\int \tan^{-1} x \cdot \ln(1 + x^2) dx$
14. Evaluate : $\int e^x \frac{1 + nx^{n-1} - x^{2n}}{(1 - x^n) \sqrt{1 - x^{2n}}} dx$





15. Evaluate : $\int \cos 2x \ln (1 + \tan x) dx$
16. Evaluate : $\int \frac{dx}{(a + b \cos x)^2}, (a > b)$
17. Evaluate : $\int \frac{\sqrt{2-x-x^2}}{x^2} dx$
18. Integrate: $\int \frac{(5x^2 - 12) dx}{(x^2 - 6x + 13)^2}$
19. If $\int \frac{3x^2 + 2x}{x^6 + 2x^5 + x^4 + 2x^3 + 2x^2 + 5} dx = F(x)$, then find the value of $[F(1) - F(0)]$, where $[.]$ represents greatest integer function.
20. Evaluate : $\int \frac{\ln (1 + \sin^2 x)}{\cos^2 x} dx.$
21. Evaluate : $\int \frac{1 + \cos \alpha \cos x}{\cos \alpha + \cos x} dx$
22. Evaluate : $\int \frac{a + b \sin x}{(b + a \sin x)^2} dx$
23. Evaluate : $\int \frac{dx}{(x - \alpha) \sqrt{(x - \alpha)(x - \beta)}}$
24. Evaluate $\int \frac{(\cos 2x)^{1/2}}{\sin x} dx$
25. Evaluate $\int \frac{\sin^3 \frac{x}{2}}{\cos \frac{x}{2} \sqrt{\cos^3 x + \cos^2 x + \cos x}} dx$
26. If $\int \frac{x^2}{x^4 + 3x^2 + 9} dx = A \tan^{-1} \left(\frac{x^2 - 3}{3x} \right) + \frac{B}{\sqrt{3}} \ln \left| \frac{x^2 - \sqrt{3}}{x^2 + \sqrt{3}} \cdot \frac{x + 3}{x - 3} \right| + c$, then find the value of $12(A + B)$.
27. Evaluate $\int \frac{3 \cos x + 2}{\sin x + 2 \cos x + 3} dx$
28. Evaluate $\int \sqrt[3]{\tan x} dx$
29. Evaluate : $\int \sqrt{\frac{\operatorname{cosec} x - \cot x}{\operatorname{cosec} x + \cot x}} \cdot \frac{\sec x}{\sqrt{1 + 2 \sec x}} dx$





Answers

1. $-\frac{1}{2} \sin 2x + C$
2. $-\left(\sin x + \frac{\sin 2x}{2}\right) + C$
3. $\frac{1}{3} \left(x + \sqrt{x^2 + 2}\right)^{3/2} - \frac{2}{\left(x + \sqrt{x^2 + 2}\right)^{1/2}} + C$
4. $\frac{\sqrt{x^2 + 2x - 3}}{8(x+1)^2} + \frac{1}{16} \cdot \cos^{-1} \left(\frac{2}{x+1}\right) + C$
5. $C - \frac{1}{3} \tan x \cdot (2 + \tan^2 x) \cdot \sqrt{4 - \cot^2 x}$
6. $\frac{2(x^2 + 1)\sqrt{x^2 + 1}}{9x^3} \cdot \left[1 - \frac{3}{2} \ln \left(1 + \frac{1}{x^2}\right)\right] + C$
7. $\frac{2(7x - 20)}{9\sqrt{7x - 10 - x^2}} + C$
8. $x; x^2 + 2x \cos \alpha + 1$
9. $\frac{1}{2} e^x [(x^2 - 1) \cos x + (x - 1)^2 \cdot \sin x] + C$
10. $e^x \left(\frac{x+1}{x^2+1}\right) + C$
11. $\frac{\sin x - x \cos x}{x \sin x + \cos x} + C$
12. $-2 \cos^4 x \cdot e^{\tan^2 x} + C$
13. $x \tan^{-1} x \cdot \ln(1 + x^2) + (\tan^{-1} x)^2 - 2x \tan^{-1} x + \ln(1 + x^2) - \left(\ln \sqrt{1 + x^2}\right)^2 + C$
14. $e^x \sqrt{\frac{1+x^n}{1-x^n}} + C$
15. $\frac{1}{2} [\sin 2x \cdot \ln(1 + \tan x) - x + \ln |\sin x + \cos x|] + C$
16. $-\frac{b \sin x}{(a^2 - b^2)(a + b \cos x)} + \frac{2a}{(a^2 - b^2)^{3/2}} \tan^{-1} \sqrt{\frac{a-b}{a+b}} \tan \frac{x}{2} + C$
17. $-\frac{\sqrt{2-x-x^2}}{x} + \frac{\sqrt{2}}{4} \ln \left| \frac{4-x+2\sqrt{2}\sqrt{2-x-x^2}}{x} \right| - \sin^{-1} \left(\frac{2x+1}{3}\right) + K$
18. $\frac{13x-159}{8(x^2-6x+13)} + \frac{53}{16} \tan^{-1} \frac{x-3}{2} + C$
19. 0
20. $\tan x \ln(1 + \sin^2 x) - 2x + \sqrt{2} \tan^{-1}(\sqrt{2} \cdot \tan x) + C$
21. $x \cos \alpha + \sin \alpha \ln \left| \frac{\cos \frac{1}{2}(\alpha - x)}{\cos \frac{1}{2}(\alpha + x)} \right| + C$
22. $-\frac{\cos x}{b + a \sin x} + C$
23. $\frac{-2}{\alpha - \beta} \cdot \sqrt{\frac{x-\beta}{x-\alpha}} + C$
24. $\sqrt{2} \log \left[\frac{\sqrt{\cot^2 x - 1} + \sqrt{2 \cot^2 x}}{\sqrt{\cot^2 x + 1}} \right] - \log [\cot x + \sqrt{\cot^2 x - 1}] + c$
25. $\sec^{-1} \left(\sqrt{\cos x} + \frac{1}{\sqrt{\cos x}} \right) + c.$
26. 5
27. $\frac{6}{5}x + \frac{3}{5} \log |\sin x + 2 \cos x + 3| - \frac{8}{5} \tan^{-1} \left(\frac{\tan \frac{x}{2} + 1}{2} \right) + C$
28. $-\frac{1}{2} \log(1 + \tan^{2/3} x) + \frac{1}{4} \log(\tan^{4/3} x - \tan^{2/3} x + 1) + \frac{\sqrt{3}}{2} \tan^{-1} \frac{2 \tan^{2/3} x - 1}{\sqrt{3}} + c$
29. $\sin^{-1} \left(\frac{1}{2} \sec^2 \frac{x}{2} \right) + C$

